

Example We consider $F = \mathbb{Q}[t]/g$ where

$$g = t^2 - t - 1 \quad \alpha := [t]_g \Rightarrow \alpha^2 - \alpha - 1 = 0.$$

Let's represent $u = [t^4 - 2t^2]_g$ as $u = x + y\alpha$
where $x, y \in \mathbb{Q}$.

$$\begin{array}{r} t^4 - 2t^2 : (t^2 - t - 1) = t^2 + t \\ \hline t^4 - t^3 - t^2 \\ \hline t^3 - t^2 - t \\ \hline t \end{array}$$

$$(t^2 + t)(t^2 - t - 1) + t = t^4 - 2t^2$$

$$\Rightarrow [t^4 - 2t^2]_g = [t]_g$$

$$\Leftrightarrow \alpha^4 - 2\alpha^2 = \alpha$$

Another way of writing: $\alpha^2 = \alpha + 1$

$$\begin{aligned}\alpha^4 - 2\alpha^2 &= \alpha^2 \cdot \alpha^2 - 2\alpha^2 \\&= (\alpha + 1) \cdot \alpha^2 - 2\alpha^2 \\&= \alpha^3 - \alpha^2 \\&= \alpha^2 \cdot \alpha - \alpha^2 \\&= (\alpha + 1) \cdot \alpha - \alpha^2 \\&= \alpha^2 + \alpha - \alpha^2 \\&= \alpha\end{aligned}$$

$$\begin{aligned}t^4 - 2t^2 &= t^2 \cdot t^2 - 2t^2 \\&= (t^2 - t - 1) \cdot t^2 + t^3 - t^2 \\&= (t^2 - t - 1) \cdot t^2 + t^2 \cdot t - t^2 \\&\quad \dots \\&= (t^2 - t - 1) \cdot (t^2 - t) + t\end{aligned}$$

How do we write α^6 as $x + \alpha y$ with $x, y \in \mathbb{Q}$?

$$\begin{aligned}\alpha^6 &= (\alpha^2)^3 = (\alpha+1)^3 = \alpha^3 + 3\alpha^2 + 3\alpha + 1 \\ &= \alpha^2(\alpha+3) + 3\alpha + 1 \\ &= (\alpha+1)(\alpha+3) + 3\alpha + 1 \\ &= \alpha^2 + 4\alpha + 3 + 3\alpha + 1 \\ &= (\alpha+1) + 4\alpha + 3 + 3\alpha + 1 \\ &= 5 + 8\alpha\end{aligned}$$

Vector spaces, linear maps and the dimension

Def A vector space V over a field K (sometimes also called a K -vector space) is a set with two operations

$$+ : V \times V \rightarrow V \quad (\text{vector addition})$$

$$\cdot : K \times V \rightarrow V \quad (\text{scalar multiplication})$$

such that:

$$\rightarrow (V, +) \text{ is an Abelian group}$$

$$\rightarrow (\alpha + \beta) \cdot u = \alpha \cdot u + \beta \cdot u$$

$$\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$$

$$1 \cdot u = u$$

$$\alpha \cdot (\beta \cdot u) = (\alpha \cdot \beta) \cdot u$$

hold for all $\alpha, \beta \in K$ and all $u, v \in V$.

Remark. For $n \in \mathbb{N}_0$ and K a field the set

$$K^n = \underbrace{K \times \dots \times K}_n \quad \text{with}$$

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) := (x_1 + y_1, \dots, x_n + y_n) \quad \text{and}$$

$$\alpha \cdot (x_1, \dots, x_n) := (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$$

for $x = (x_1, \dots, x_n) \in K^n$, $y = (y_1, \dots, y_n) \in K^n$ and $\alpha \in K$

is a vector space over K .

Def Let K be a field and V and W be vector spaces over K .

Then a map $F: V \rightarrow W$ is called K -linear (or just

linear) if $F(\alpha u + \beta v) = \alpha F(u) + \beta F(v)$

holds for all $\alpha, \beta \in K$ and $u, v \in V$.

Def Let V be a vector space over a field K .

If there exists a bijective K -linear map

$F: K^n \rightarrow V$ for $n \in \mathbb{N}_0$, then V is said to be n -dimensional over K .

Remark. This n is uniquely determined by V and is called the dimension of V .

Notation: $\dim_K(V)$ or merely $\dim(V)$ if the choice of K is clear from the context.

Proposition 27 Let K be a field and $g \in K[t]$ be a polynomial of degree $n \in \mathbb{N}$. Then $R := K[t]/g$ is a K -vector space of dimension n .

Sketch of the proof:

There is an addition in R , the addition of cosets, and there is a scalar multiplication,

$$\alpha \cdot [f]_g := [\alpha f]_g \quad (\alpha \in K, f \in K[t])$$

It's straight forward to see that R is a vector space w.r.t. these two operations.

Let $\omega := [t]_g$. It turns out that the map

$\varphi: K^n \rightarrow R$ given by

$$\varphi(c_0, \dots, c_{n-1}) := \omega \cdot \omega^0 + \dots + c_{n-1} \cdot \omega^{n-1}$$

is K -linear and bijective; it's quite straightforward to show this. $\square$

Example Let's construct a finite field with 4 elements.

We take $\mathbb{Z}_2$, the finite field with 2 elements, and construct $F = \mathbb{Z}_2[t]/g$ where $g \in \mathbb{Z}_2[t]$ is an irreducible polynomial of degree 2.

$$\left. \begin{array}{l} t^2 = t \cdot t \\ t^2 + 1 \\ t^2 + t = t \cdot (t+1) \\ t^2 + t + 1 \end{array} \right\} \begin{array}{l} \text{all polynomials of degree 2} \\ \text{in } \mathbb{Z}_2[t] \end{array}$$

$t^2 + 1$ has the zero $t = 1$ in $\mathbb{Z}_2$:

$$1^2 + 1 = 1 + 1 = 0 \text{ in } \mathbb{Z}_2$$

$$\Rightarrow t^2 + 1 = (t+1)^2 \text{ in } \mathbb{Z}_2[t], \text{ why so?}$$

$$(t+1)^2 = (t+1)(t+1) = t^2 + t + t + 1 = t^2 + \underbrace{(1+1)}_{0 \text{ in } \mathbb{F}_2} \cdot t + 1$$

What about $t^2 + t + 1$?

It is irreducible. Indeed, if it were reducible then it would be a product of two polynomials of degree one. As polynomials of degree one in $\mathbb{F}_2[t]$ have a zero in $\mathbb{F}_2$, $t^2 + t + 1$ too would have a zero in $\mathbb{F}_2$. But it doesn't:

$$0^2 + 0 + 1 = 1 \neq 0$$

$$1^2 + 1 + 1 = 1 + 1 + 1 = 0 + 1 = 1 \neq 0$$

$\Rightarrow F = \mathbb{F}_2[t]/g$ with $g = t^2 + t + 1$ is a field.

Let us denote $[t]_g$ as α . $\alpha := [t]_g$.

$$\alpha^2 + \alpha + 1 = 0 \iff \alpha^2 = -\alpha - 1$$

$\Rightarrow F = \{ \underbrace{0, 1}_{\mathbb{Z}_2}, \alpha, 1+\alpha \}$ and it contains $\mathbb{Z}_2$

F is a two-dimensional vector space over $\mathbb{Z}_2$.

Let's create tables for $+$ and $\cdot$ in F .

$+$	0	1	α	$1+\alpha$
0	0	1	α	$1+\alpha$
1	1	0	$1+\alpha$	α
α	α	$1+\alpha$	0	1
$1+\alpha$	$1+\alpha$	α	1	0

$\cdot$	0	1	α	$1+\alpha$
0	0	0	0	0
1	0	1	α	$1+\alpha$
α	0	α	$1+\alpha$	1
$1+\alpha$	0	$1+\alpha$	1	α

$$\alpha(1+\alpha) = \alpha^2 + \alpha$$

$$= 1 + \alpha + \alpha = 1$$

$$(1+\alpha)^2 = 1 + (1+1)\alpha + \alpha^2$$

$$= 1 + \alpha^2 = \alpha$$

Example Let's determine all degree-three irreducible polynomials in $\mathbb{Z}_2[t]$. Let's consider an $f \in \mathbb{Z}_2[t]$ of $\deg f = 3$. When is it irreducible? It turns out that in the case $\deg f = 3$, f is irreducible if and only if f has no zeros in $\mathbb{Z}_2$ (which means, $f(0) \neq 0$ and $f(1) \neq 0$). Indeed, if f has a zero $a \in \mathbb{Z}_2$, then f is divisible by $t-a$ and thus is a product of a degree-one and degree-two polynomial. This shows that once f is reducible. Conversely, if f is reducible, since $\deg f = 3$, it factors into the product of polynomials of degree 1 and degree 2. The factor of degree one has a zero, which is also a zero of f .

So, f is reducible if and only if f has a zero in $\mathbb{Z}_2$.
Consequently, f is irreducible if and only if f has no zeros in $\mathbb{Z}_2$.

So, an irreducible f has the form

$$f = t^3 + \underbrace{a}_{\in \mathbb{Z}_2} t^2 + \underbrace{b}_{\in \mathbb{Z}_2} t + 1$$

$\hookrightarrow$ makes sure that $f(0) \neq 0$.

The condition $f(0) \neq 0$ is guaranteed for this f .

The only choices with $f(1) \neq 0$ are

$a=1, b=0$ and $a=0, b=1$. So, there are exactly

two irreducible polynomials of degree 3 in $\mathbb{Z}_2[t]$:

They are $t^3 + t^2 + 1$ and $t^3 + t + 1$.

Example (follow up on the previous one) We've got two fields with 8 elements each:

$F := \mathbb{Z}_2[t]/f$, where $f := t^3 + t^2 + 1$, and

$G := \mathbb{Z}_2[t]/g$, where $g := t^3 + t + 1$.

let's determine $f(t+1)$:

$$\begin{aligned} f(t+1) &= (t+1)^3 + (t+1)^2 + 1 \\ &= (t+1)^3 + (t+1)(t+1) + 1 \\ &= (t+1)^3 + t^2 + \underbrace{(1+1) \cdot t}_{=0} + 1 + 1 \\ &= (t+1)^3 + t^2 + 1 + 1 \\ &= (t+1)^3 + t^2 \\ &= (t+1)^2 \cdot (t+1) + t^2 \\ &= (t^2+1) \cdot (t+1) + t^2 \\ &= t^3 + t^2 + t + 1 + t^2 \\ &= t^3 + t + 1 = g(t). \end{aligned}$$

This means that for the coxets in $\mathbb{Z}_2[t]/f$ and
in $\mathbb{Z}_2[t]/g$ there is a natural correspondence:
for $h \in \mathbb{Z}_2[t]$ we have

$$[h(t)]_{f(t)} \iff [h(t+1)]_{f(t+1)} = [h(t+1)]_{g(t)}.$$

This correspondence turns out to be a unitary isomorphism. This shows that the fields F and G are essentially the same thing. Let's make this abstract message more explicit.

Let $\alpha := [t]_f$, which means $f(\alpha) = \alpha^3 + \alpha^2 + 1 = 0$.

What is $g(\alpha+1)$? We know $f(t+1) = g(t)$.

So $g(\alpha+1) = f((\alpha+1)+1) = f(\alpha) = 0$. So,

$\alpha+1$ is a zero of $g(t) = t^3 + t + 1$.

But in $G = \mathbb{F}_2[t]/g$ one has the coset $\beta := [t]_g$,

which satisfies $g(\beta) = \beta^3 + \beta + 1 = 0$.

So, $\alpha+1 \in F \iff \beta \in G$.

Example Let's invert some elements in $\mathbb{Z}_2[t]/g$, where
 $g = t^3 + t + 1 \in \mathbb{Z}_2[t]$. Let's use
 $\beta = [t]_g$, which means $\beta^3 + \beta + 1 = 0$.

Let's try to invert $\beta^2 + \beta + 1$. Let's consider the polynomials

$$f = t^2 + t + 1, \quad (1)$$

$$g = t^3 + t + 1. \quad (2)$$

We run the EEA on the input (f, g) .

$$\begin{array}{r} (t^3 + t + 1) : (t^2 + t + 1) = t + 1 \\ \underline{t^3 + t^2 + t} \\ t^2 + 1 \\ \underline{t^2 + t + 1} \\ t \end{array}$$

We multiply (1) with $t + 1$ and add the result to (2).
 This gives:

$$\begin{cases} f = t^2 + t + 1 & (3) \\ (t+1)f + g = t & (4) \end{cases}$$

$$\boxed{\begin{array}{r} (t^2 + t + 1) : t = t + 1 \\ \underline{t^2} \\ t + 1 \\ \underline{t} \\ 1 \end{array}}$$

We multiply (4) with $t+1$ and add the result to (3)

$$\begin{cases} t^2 f + (t+1)g = 1 & (5) \\ (t+1)f + g = t & (6) \end{cases}$$

(5) is enough to get the inverse. (5) implies

$$t^2 \cdot f \equiv 1 \pmod{g}.$$

This means

$$t^2 \cdot (t^2 + t + 1) \equiv 1 \pmod{g},$$

which means that $\beta^2 \cdot (\beta^2 + \beta + 1) = 1$
in $\mathbb{F}_2[t]/q$. So, $(\beta^2 + \beta + 1)^{-1} = \beta^2$.

Def Two fields F and G are called isomorphic if they are isomorphic as unitary commutative rings;

That means, there exists a unitary isomorphism

$$\varphi: F \rightarrow G.$$

Theorem 28 (The fundamental theorem of finite fields)

The size of every finite field is a power $q = p^n$

of a prime number p (here, $n \in \mathbb{N}$).